

Operation: RemoveItem(itemId: Integer)

Cross References: Use Cases: Update Inventory and Alert on Shortages

Preconditions: An Item instance with the matching itemId exists in the system.

Postconditions:

- The Item instance was deleted (instance deletion).
- The association between the Item instance and its corresponding ItemType was broken (association broken).
- The total quantity attribute of the ItemType was decreased by 1 (attribute modification).
- If the ItemType's quantity dropped below its defined minimum stock level:

An Alert instance a was created (instance creation).

a was associated with the ItemType (association formed).